Indian Institute of Technology (IIT-Bombay)

AUTUMN Semester, 2026
COMPUTER SCIENCE AND ENGINEERING

CS230: Digital Logic Design and Computer Architecture

Tutorial - I

Full Marks: 0

Time allowed: ∞ hours

1. A new flip-flop, called an MN flip-flop, is constructed from a JK flip-flop as follows:

$$J = M, \qquad K = N \oplus M$$

(a) Derive the state-transition table and excitation table for this new flip-flop.
(b) Derive the characteristic equation.
(c) Construct a D flip-flop from this new flip-flop.

Solution

(a) State-Transition Table and Excitation Table

JK Flip-Flop Characteristic Table

J	K	Q_{t+1}
0	0	Q_t
0	1	0
1	0	1
1	1	$\overline{Q_t}$

MN Flip-Flop State Transition Table

Using:

$$J = M, \qquad K = M \oplus N$$

M	N	Q_t	J	K	Q_{t+1}
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	1	0
0	1	1	0	1	0
1	0	0	1	1	1
1	0	1	1	1	0
1	1	0	1	0	1
1	1	1	1	0	1

Excitation Table

Q_t	Q_{t+1}	Required (M, N)
0	0	$(0,0),(0,1)$
0	1	$(1,0),(1,1)$
1	0	$(0,1),(1,0)$
1	1	$(0,0),(1,1)$

(b) Characteristic Equation

$Q_t \backslash MN$	00	01	11	10
0	0	0	1	1
1	1	0	1	0

Minterms where $Q_{t+1} = 1$:

- $Q_t = 0, MN = 11 \Rightarrow M \cdot N \cdot \overline{Q_t}$
- $Q_t = 0, MN = 10 \Rightarrow M \cdot \overline{N} \cdot \overline{Q_t}$
- $Q_t = 1, MN = 00 \Rightarrow \overline{M} \cdot \overline{N} \cdot Q_t$
- $Q_t = 1, MN = 11 \Rightarrow M \cdot N \cdot Q_t$

From the state transition table, we derive the characteristic equation:

$$Q_{t+1} = \left(\overline{M}\,\overline{N}\,Q_t\right) + \left(M\,\overline{N}\,\overline{Q_t}\right) + \left(M\,N\,\overline{Q_t}\right) + (M\,N\,Q_t)$$

Simplifying:

$$Q_{t+1} = \overline{M}\,\overline{N}\,Q_t + M\,\overline{N}\,\overline{Q_t} + MN(\overline{Q_t} + Q_t)$$

Since:

$$\overline{Q_t} + Q_t = 1$$

we get:

$$\boxed{Q_{t+1} = M\overline{N}\,\overline{Q_t} + \overline{M}\,\overline{N}\,Q_t + MN}$$

Alternatively,

$$\boxed{Q_{t+1} = N\overline{Q_t} + \overline{N}(\overline{M}Q_t + M\overline{Q_t})}$$

(c) Construct a D Flip-Flop from MN Flip-Flop

We want the MN flip-flop to behave like a D flip-flop:

$$Q^+ = D$$

Choose:

$$M = D, \qquad N = Q$$

Then:

$$J = D, \qquad K = D \oplus Q$$

This results in the desired D flip-flop behavior.

2. Construct a counter for the following sequence using T flip-flops: 000 → 010 → 111 → 100 → 011. (a) Draw the state-table (b) Construct the next-state map (c) Derive the inputs of the T flip-flops (d) Draw the final circuit

Solution:

(a) State Table

Present State ($Q_2Q_1Q_0$)	Next State ($Q_2^+Q_1^+Q_0^+$)
000	010
010	111
111	100
100	011
011	000

—

(b) Next-State Map

Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+
0	0	0	0	1	0
0	1	0	1	1	1
1	1	1	1	0	0
1	0	0	0	1	1
0	1	1	0	0	0

—

(c) Derive the T Flip-Flop Inputs

Recall the excitation table for T flip-flops:

Q	Q^+	T
0	0	0
0	1	1
1	0	1
1	1	0

Thus,

$$T = Q \oplus Q^+$$

Calculate the T inputs for each flip-flop:

Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+	$T_2 = Q_2 \oplus Q_2^+$	$T_1 = Q_1 \oplus Q_1^+$	$T_0 = Q_0 \oplus Q_0^+$
0	0	0	0	1	0	0	1	0
0	1	0	1	1	1	1	0	1
1	1	1	1	0	0	0	1	1
1	0	0	0	1	1	1	1	1
0	1	1	0	0	0	0	1	1

—

Simplify the T Inputs

For T_2:

$$T_2 = \sum m(2, 4)$$

Where:

$$m_2 = Q_2' Q_1 Q_0', \quad m_4 = Q_2 Q_1' Q_0'$$

Thus,

$$T_2 = Q_2' Q_1 Q_0' + Q_2 Q_1' Q_0'$$

—

For T_1: T_1 is 1 for all states except at 010.
Thus,

$$T_1 = \overline{Q_2' Q_1 Q_0'} = Q_2 + Q_1' + Q_0$$

—

For T_0: T_0 is 0 only at 000, so

$$T_0 = Q_2 + Q_1 + Q_0$$

—

(d) Final Circuit

The final circuit consists of three T flip-flops with inputs:

$$\boxed{\begin{cases} T_2 = Q_2' Q_1 Q_0' + Q_2 Q_1' Q_0' \\ T_1 = Q_2 + Q_1' + Q_0 \\ T_0 = Q_2 + Q_1 + Q_0 \end{cases}}$$

Logic gates needed:

- NOT gates for Q_2', Q_1', Q_0'
- AND gates for the two terms in T_2
- OR gates to combine terms in T_2, and to generate T_1 and T_0
- Connect T_i inputs to corresponding T flip-flops Q_i

All flip-flops are triggered by the common clock signal.

3. A sequential circuit has one flip flop Q, two inputs X and Y and one output S. The circuit consists of a full subtractor circuit connected to a D flip flop as shown in Figure 1 below. Derive the state table and state diagram for the sequential circuit.

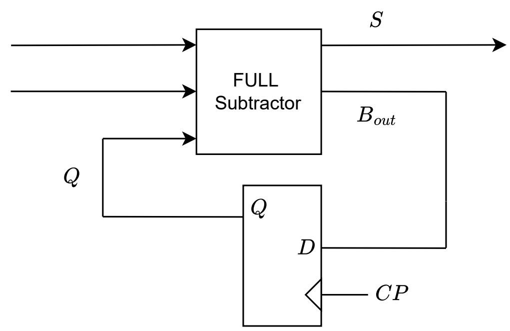


Figure 1

Solution total 8 rows

Q (Present State)	x	y	B_{out} = Q(t+1)	S = x ⊕ y ⊕ Q
0	0	0	0	0
0	0	1	1	1
0	1	0	0	1
0	1	1	0	0
1	0	0	1	1
1	0	1	1	0
1	1	0	0	0
1	1	1	1	1

Table 1: State Table of Sequential Circuit with Full Subtractor

4. In this question we construct a serial BCD to Excess-3 code converter from serial input. In other words, the input will be provided, and the output will be generated one bit at a time. Excess-3 code can be generated by adding $(0011)_2$ with the BCD code. Therefore, you need to consider 4 bits of input at a time as a valid BCD code. The expected input and outputs provided in Figure 2. Observe that the inputs will be provided from LSB to MSB and the outputs will be generated in the same manner (more precisely, the LSB of the input will be processed first and the LSB of the output will be generated first). So you have to consider each entry in the table from right to left while processing.

- In this question you will be generating a circuit for this state machine. First generate the state-transition graph and the state-transition table. *Hint:* The state-transition graph will have the states as the nodes and transitions of the form input/output as the links. The number of states can be quite large in this case. To keep the number of states low, we use some tricks.
  - **Observation 1:** Since the input is of 4 bits, you need to transit through 4 states for each input.
  - **Observation 2:** Let S_t be any state processing the i-th bit of input for some string. It will have two possible transitions out of it for $x_i = 0$ and $x_i = 1$. Observe (from the table) that, for any bit location i, the output $y_i = \overline{x_i}$ or $y_i = x_i$ for $x_i \in \{0, 1\}$. In other words, the two transitions from any given state will be either (0/1), (1/0) or (0/0), (1/1). There will be no other transitions, such as, (0/1), (1/1).
  - **Observation 3:** There can be total 16 possible states. But many of these states will be equivalent and, therefore, can be merged together.
- **Trick 1:** There will some state transitions which are impossible. Find them out and add dummy transitions for them keeping observation 2 in mind.
- **Trick 2:** The initial state S_0 will also be the final state.
- **Trick 3:** Eliminate the equivalent states. Two states S_i and S_j are equivalent if and only if for every possible input sequence, the same output sequence is produced, regardless of whether S_i or S_j is the starting state. This much should be sufficient for you to solve the problem.

(b) Perform the state assignment using binary encoding. If you have any unused state, use it as don't care.

Decimal Digit	8-4-2-1 Code (BCD)	Excess-3 Code
0	0000	0011
1	0001	0100
2	0010	0101
3	0011	0110
4	0100	0111
5	0101	1000
6	0110	1001
7	0111	1010
8	1000	1011
9	1001	1100

Figure 2: BCD to Excess-3 Code Converter

solution

PS	X = 0	X = 1
S0	S1,1	S2,0
S1	S3,1	S4,0
S2	S5,0	S6,1
S3	S7,0	S8,1
S4	S9,1	S10,0
S5	S11,1	S12,0
S6	S13,1	S14,0
S7	S0,0	S0,1
S8	S0,0	-
S9	S0,0	-
S10	S0,1	-
S11	S0,0	S0,1
S12	S0,1	-
S13	S0,0	-
S14	S0,1	-

Figure 3: State Table

PS	X = 0	X = 1
S0	S1,1	S2,0
S1	S3,1	S4,0
S2	S4,0	S4,1
S3	S7,0	S7,1
S4	S7,1	S10,0
S5	S7,1	S10,0
S6	S7,1	S10,0
S7	S0,0	S0,1
S8	S0,0	-
S9	S0,0	-
S10	S0,1	S0,0
S11	S0,0	S0,1
S12	S0,1	-
S13	S0,0	-
S14	S0,1	-

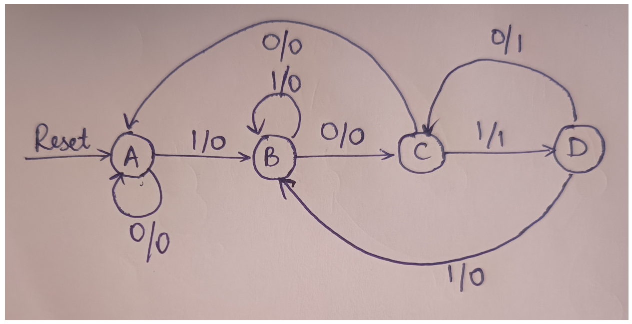
Figure 4: Minimizing State Table

PS	X = 0	X = 1
S0	S1,1	S2,0
S1	S3,1	S4,0
S2	S4,0	S4,1
S3	S7,0	S7,1
S4	S7,1	S10,0
S7	S0,0	S0,1
S10	S0,1	S0,0

Figure 5: Minimized State Table

5. Design a circuit that detects the signal 1010 or 101 in a sequence of bits. For example, when given a sequence of 101011011110001 as an input, the output has to be 001110010000000. Use any flip-flop possible.

solution



Defining the states as follows:

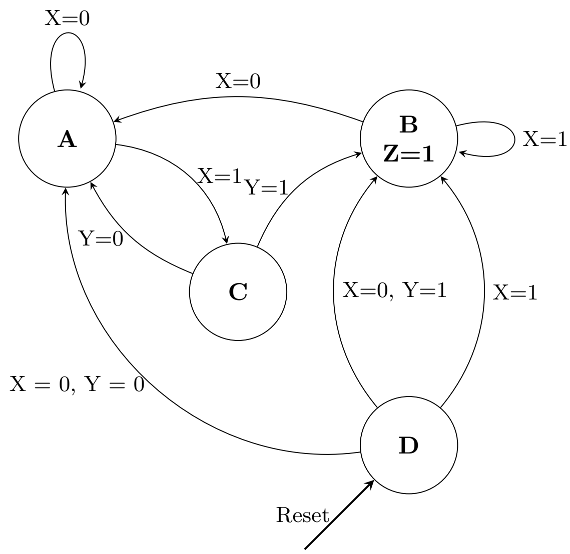
State	Encoding (MN)
A	00
B	01
C	11
D	10

The state transition table is as follows:

M	N	x	M_	N_	z
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	1	0	0
0	1	1	0	1	0
1	0	0	0	0	0
1	0	1	1	1	1
1	1	0	1	0	1
1	1	1	0	1	0

$$M_{\_} = M'Nx' + MN'x + MNx'$$
$$N_{\_} = x$$
$$z = MN'x + MNx'$$

6. Design the circuit for the following finite state machine with inputs X and Y. Define a method to represent the states in the circuit



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solution

This can be solved with Karnaugh maps with the data

Considering the states as follows:

State	Encoding (MN)
A	00
B	01
C	11
D	10

The state transition table is as follows:

Previous State	X	Y	Next State	Z
A	0	0	A	0
A	0	1	A	0
A	1	0	C	0
A	1	1	C	0
B	0	0	A	0
B	0	1	A	0
B	1	0	B	1
B	1	1	B	1
C	0	0	A	0
C	0	1	B	1
C	1	0	A	0
C	1	1	B	1
D	0	0	A	0
D	0	1	B	1
D	1	0	B	1
D	1	1	B	1

or using encoding way

MN	X	Y	M^+N^+	Z
00	0	0	00	0
00	0	1	00	0
00	1	0	11	0
00	1	1	11	0
01	0	0	00	0
01	0	1	00	0
01	1	0	01	1
01	1	1	01	1
11	0	0	00	0
11	0	1	00	0
11	1	0	01	1
11	1	1	01	1
10	0	0	00	0
10	0	1	01	1
10	1	0	01	1
10	1	1	01	1

Table 2: State transition/output table

Using two D Flip Flops, one for M+ and another for N+. The K-Maps for M+, N+, z is given in the fig- 6

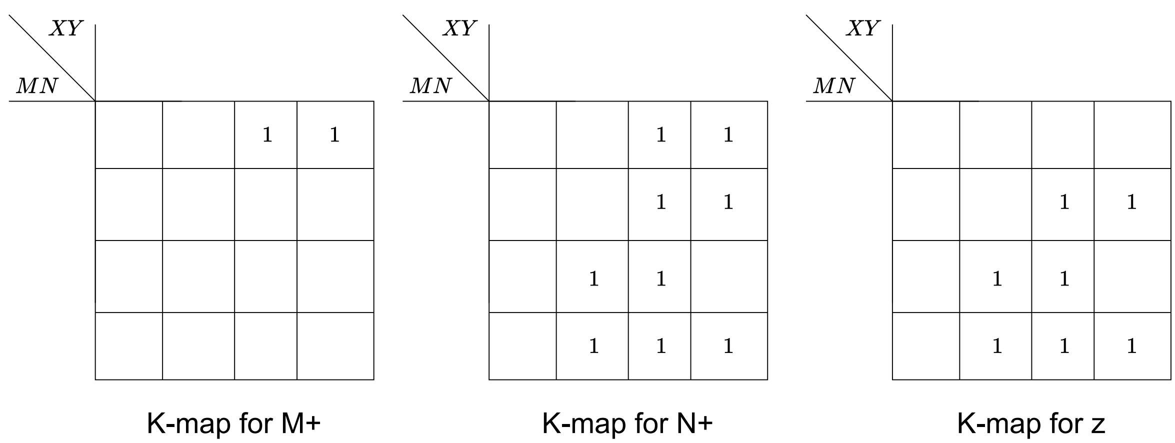


Figure 6: K-Maps for M+, N+, z

Using the K-Maps, we get,

$$M^+ = M'N'X$$
$$N^+ = M'X + MY + MN'X$$
$$Z = MY + (M \oplus N)X$$

The final circuit Diagram looks like this

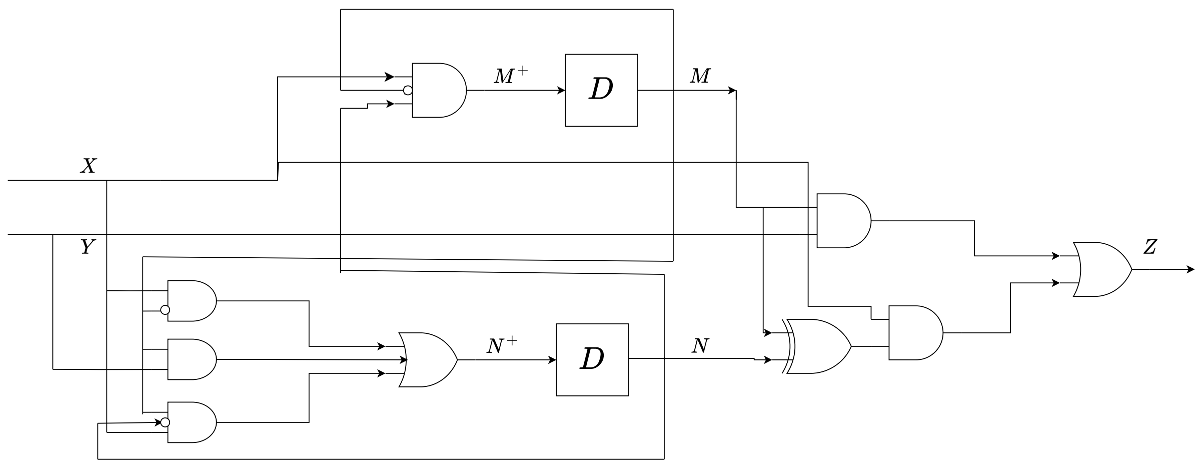
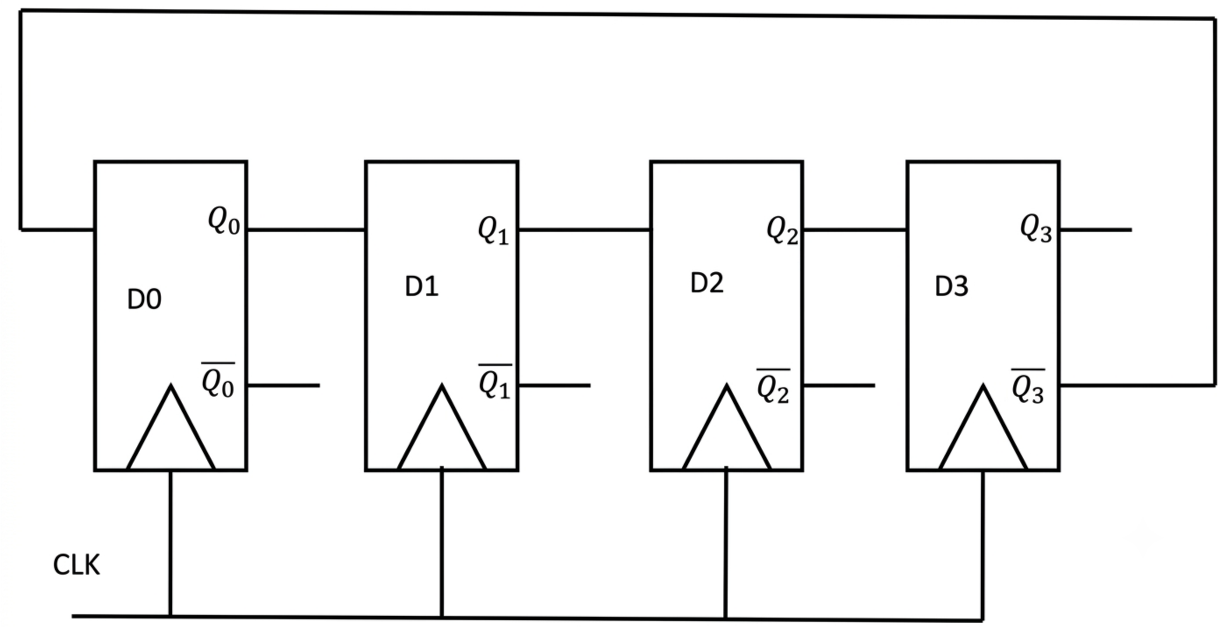


Figure 7: Final circuit

7. Consider the given sequential circuit designed using D-Flip-flops. The circuit is initialized with some value (initial state). The number of distinct states the circuit will go through before returning back to the initial state is (Answer in integer excluding initial state and Assume initial state value as 0)



Solution

Characteristic of D Flip-Flop:

$$Q_{\text{new}} = \text{Input}$$

Let the output sequence be $Q_0Q_1Q_2Q_3$:

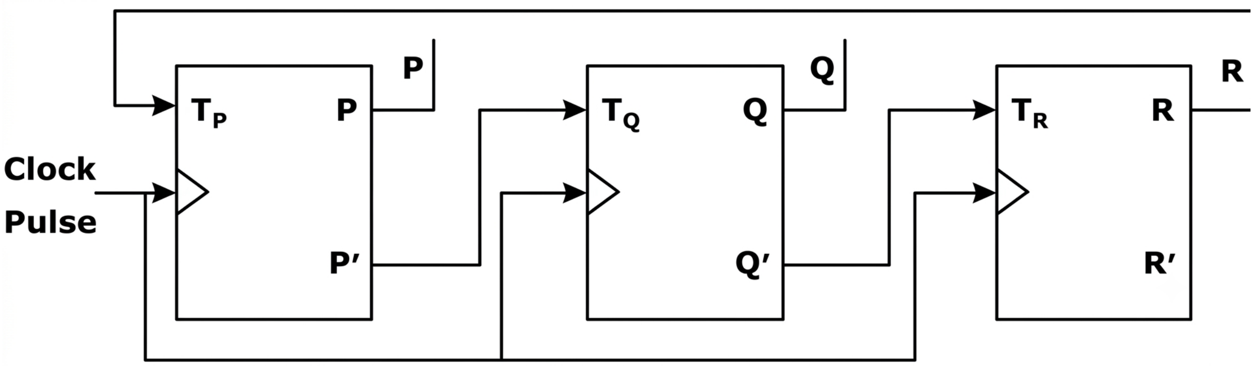
- The input of D_0 flip-flop is $\overline{Q}_3$. Hence, when a clock is applied, the output of D_0 flip-flop is $Q_0^+ = \overline{Q}_3$.
- The input of D_1 flip-flop is Q_0. Hence, when a clock is applied, the output of D_1 flip-flop is $Q_1^+ = Q_0$.
- The input of D_2 flip-flop is Q_1. Hence, when a clock is applied, the output of D_2 flip-flop is $Q_2^+ = Q_1$.
- The input of D_3 flip-flop is Q_2. Hence, when a clock is applied, the output of D_3 flip-flop is $Q_3^+ = Q_2$.

Assume initial state value as 0000:

Clock Pulse	Q_0	Q_1	Q_2	Q_3	**Decimal Value**
Initial	0	0	0	0	0
1	1	0	0	0	8
2	1	1	0	0	12
3	1	1	1	0	14
4	1	1	1	1	15
5	0	1	1	1	7
6	0	0	1	1	3
7	0	0	0	1	1
8	0	0	0	0	0

The number of distinct states the circuit will go through before returning back to the initial state (excluding the initial state) is **7**.

8. Consider a 3-bit counter, designed using T flip-flops, as shown below :



Assuming the initial state of the counter given by PQR as 000, what are the next three states?

Solution From the given 3-state counter made from T flip-flops, the next input sequence is as follows:

$$T_P = R, \quad T_Q = \overline{P}, \quad T_R = \overline{Q}$$

Initial State			Current Input			Next State		
P	Q	R	T_P	T_Q	T_R	P^+	Q^+	R^+
0	0	0	0	1	1	0	1	1
0	1	1	1	1	0	1	0	1
1	0	1	1	0	1	0	0	0

In a T flip-flop, for low input (0), the next state is Q_n (current state), and for high input (1), it toggles/complements the present state ($\overline{Q}_n$).

$$011 \rightarrow 101 \rightarrow 000$$